

Executive Summary

Locked Gravitational Saturation Cosmology

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Purpose

This manuscript presents a complete effective cosmological model in which gravity possesses a finite organizational or saturation response. The model is constructed to recover ordinary Friedmann evolution at low density, replace the initial singularity with a finite-density bounce, and produce a late residual with equation of state close to -1 without adding a separate dark-energy fluid.

Core construction

- A modified high-density Friedmann constraint, $H^2 = (\rho/3)(1-\rho/\rho_s)$, with a locked saturation density.
- Two canonical scalar fields, C and phi, representing organizational and phase sectors.
- A stable potential minimum with residual floor V_{∞} .
- A one-sided contraction-era transfer that converts excess scalar kinetic energy into radiation while preserving total conservation.
- A regular cosmic-time or affine evolution through $H = 0$, followed by a locked expanding branch.
- A modified CAMB implementation carrying the same locked background into perturbation and observable calculations.

Reported locked results

Quantity	Reported value
100 theta*	1.041033
Drag horizon r_drag	149.960 Mpc
sigma8	0.779680
S8	0.79781
Planck 2018 lensing	chi-square = 10.7379 for 9 bins; p = 0.294
Late field equation of state	w_C(0) approximately -0.99988

What is being requested

The author is seeking independent technical assessment, not endorsement. Reviewers are asked to test the mathematical consistency, physical assumptions, numerical implementation, reproducibility, and observational interpretation, and to identify errors, missing literature, or reasons the model may not be suitable for peer-reviewed publication.

Scope and limitations

The work is an effective model. A unique derivation from a complete microscopic or covariant fundamental theory is not claimed. A full primary-CMB likelihood analysis with nuisance parameters, independent code reproduction, and nonlinear structure-formation studies remain necessary before broad physical conclusions can be drawn.